

Towards Trustworthy Human-AI Coexistence: A Physical-Social Benchmark for Safe Embodied AI in Symmetrical Reality

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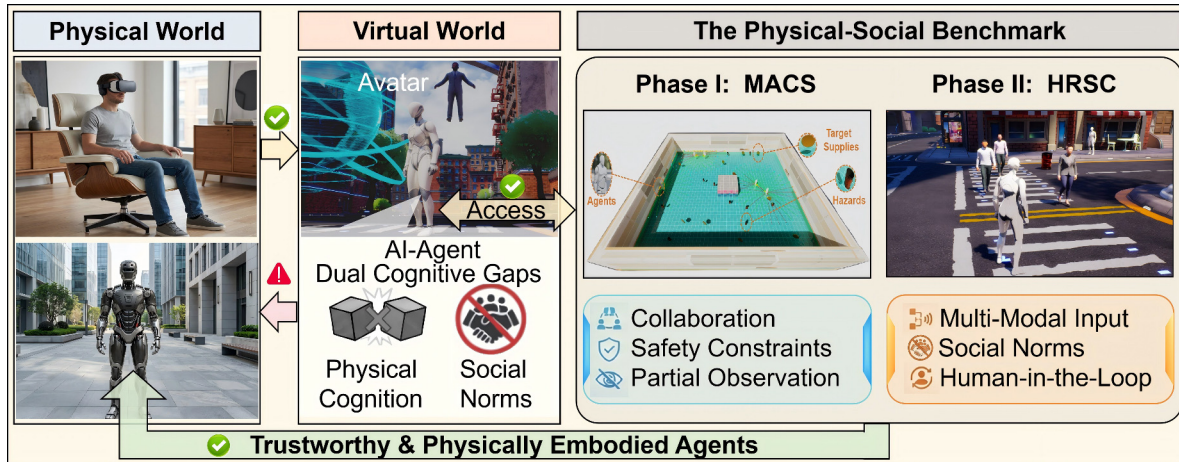


Figure 1: Bridging the Gap: From Virtual-Native Agents to Trustworthy Embodiment. While humans easily access virtual worlds, AI agents suffer from a lack of physical and social cognition. Physical-Social Benchmark functions as an access verification mechanism, filtering unsafe behaviors through rigorous physical constraints and social norm evaluations, enabling safe human-machine coexistence in the physical world.

ABSTRACT

Symmetrical Reality envisions seamless human-AI coexistence, yet virtual-native AI agents lack the physical cognition and social understanding required for safe physical embodiment. We identify these deficiencies as “Dual Cognitive Gaps.” To bridge this asymmetry, we propose the Physical-Social Benchmark, a simulation benchmark serving as a mandatory access verification mechanism. It evaluates agents through a curriculum fusing immutable physical laws with human-centric social norms. Preliminary experimental results demonstrate that our framework significantly reduces dangerous behaviors and fosters emergent collaboration. Furthermore, our user study ($N = 12$) confirms that socially compliant agents receive significantly higher favorability ratings ($p = 0.012$). These findings establish a foundation for trustworthy human-AI coexistence in Symmetrical Reality.

Index Terms: Symmetrical Reality, Human-AI Coexistence, Embodied AI

1 INTRODUCTION

The emergence of Symmetrical Reality [4] envisions a unified continuum for human-AI coexistence. However, this integration is hindered by a significant “Access Asymmetry.” While humans safely explore virtual worlds via AR/VR, granting virtual-native AI agents embodied access to the physical world introduces critical risks. We identify that these agents suffer from “Dual Cognitive Gaps”: they are deficient in both physical cognition (e.g., collision avoidance)

and social norm understanding (e.g., proxemics). Deploying such agents without rigorous verification is inherently unsafe [2].

To bridge these gaps, we propose the “Physical-Social Benchmark,” a simulation framework embedded with physical and social constraints. Serving as the cornerstone of Symmetrical Reality, this verification mechanism ensures that only trustworthy agents traverse the virtual-physical divide by passing a two-stage curriculum:

Phase I: Basic Physical Safety Verification. Within a simplified Multi-Agent Collaborative Search (MACS) benchmark, agents must learn to adhere to rigid physical constraints (e.g., collision avoidance, thrust penalty) and demonstrate basic multi-agent collaboration under partially observable conditions.

Phase II: High-Fidelity Social Coexistence. Progressing to the Human-Robot Social Coexistence (HRSC) stage, we introduce intent-driven human avatars to simulate realistic social dynamics. In this context, humans also serve as evaluators. We argue that the immersion afforded by VR is essential for eliciting visceral psychological responses—such as fear or avoidance—to social aggression exhibited by agents, thereby capturing the authentic feedback necessary for fostering genuine human-AI trust. Here, agents must master socially compliant navigation—strictly adhering to proxemics and kinematic smoothness to minimize intrusion while executing complex tasks with multi-modal inputs.

2 METHODOLOGY

We propose a rigorous two-phase curriculum designed to assess embodiment readiness prior to physical deployment.

Phase I: Basic Safety Verification. Developed within Unreal Engine 5, the MACS benchmark simulates a partially observable post-flood rescue scenario populated with stochastic dynamic hazards and static obstacles (Fig. 2). Our evaluation rests on three pillars: synchronous multi-agent collaboration (where successful collection requires simultaneous contact by $N \geq 2$ agents), rigid safety con-

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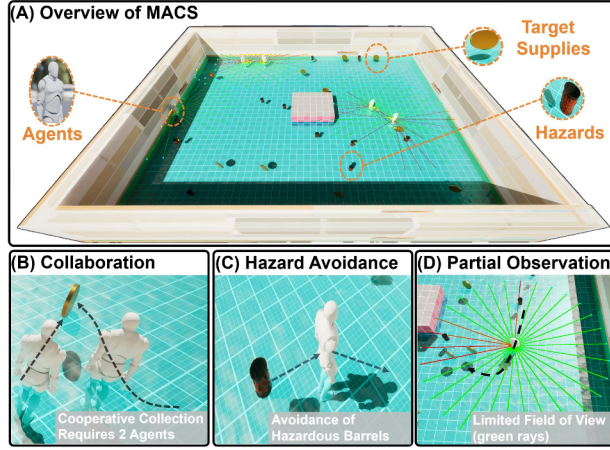


Figure 2: Visualization of the Multi-Agent Cooperative Search (MACS) Task Scenarios.

straints, and energy-efficient navigation under partial observability. Technically, the system employs a continuous 2D action space and sparse radial sensor arrays (providing distance, bearing, and velocity) within a highly parameterizable configuration space (Table 1). Crucially, we introduce a mixing coefficient λ to balance individual versus collective objectives: $R_{total}^{(i)} = \lambda R_{local}^{(i)} + (1 - \lambda) R_{global}$. Systematic variation of these configurations allows for the elicitation and analysis of diverse emergent behaviors.

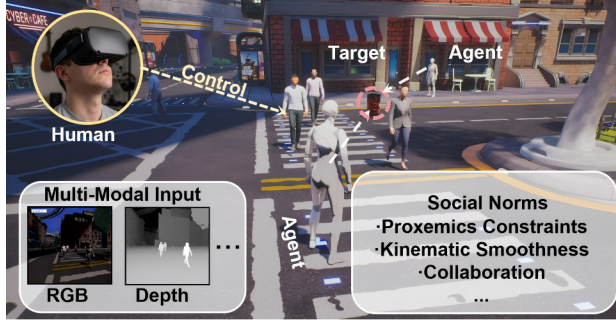


Figure 3: Visualization of the Human-Robot Social Coexistence (HRSC) Task Scenarios.

Phase II: Social Coexistence Assessment. The HRSC framework is deployed within a high-fidelity urban simulation featuring dense, hybrid-driven crowds (integrating heuristic models and Human-in-the-Loop interaction) to rigorously evaluate social compliance (Fig. 3). Agents leverage multimodal perception (RGB-D, LiDAR, GPS) to execute collaborative complex tasks under strict social constraints. Performance is quantified via a composite metric integrating task efficacy with subjective human trust scores.

The framework is implemented, featuring high-fidelity environments, naturalistic crowd behaviors, and multimodal pipelines. A key innovation is the integration of a Human-in-the-Loop. Validation is conducted using keyboard, with a planned transition to VR headsets for immersive evaluation of social compliance.

3 PRELIMINARY EXPERIMENTAL RESULTS

For Phase I, we deployed MAPPO [3], IPPO [1], and a random policy. Quantitative results (Table 2) show that MAPPO achieved a mean return of 16.40, significantly outperforming IPPO (10.02). Qualitatively, two key emergent behaviors highlight the benchmark’s ability to foster embodied cognition: (1) Safety: Agents

Table 1: Detailed Environment Specifications and Default Hyperparameters for the MACS Task.

Parameter	Description	Default
n_agents	Number of rescue agents	5
n_supplies	Number of valuable supply items	10
n_hazards	Number of hazard items	5
n_coop	Agents required for successful cooperation	2
n_sensors	Number of sensors per agent	30
sensor_range	Maximum sensing range	500.0
max_cycles	Maximum steps per episode	500
supply_reward	Reward for successfully collecting supplies	10.0
hazard_reward	Penalty for encountering hazards	-2.0
encounter_reward	Reward for touching a supply without capture	0.01
thrust_penalty	Energy cost multiplier per movement step	-0.01
local_ratio (λ)	Ratio of local rewards to global rewards	0.9

Table 2: Performance comparison of baseline algorithms on the MACS task. Metrics reported are the average hazard collisions (per agent per episode) and the mean episode return (per agent).

Method	Avg. Hazard Collisions ($\downarrow$)	Mean Ep. Return ($\uparrow$)
MAPPO	1.26	16.40
IPPO	3.65	10.02
Random	5.16	-9.65

demonstrated proactive obstacle avoidance. Specifically, MAPPO achieved a lower average collisions (1.26) compared to the random baseline (5.16), indicating effective internalization of physical constraints. (2) Collaboration: Agents exhibited cohesive movement patterns that directly enhanced item collection efficiency.

In Phase II, we conducted a preliminary user study ($N = 12$) evaluating the Social Acceptability of the agents, using a 5-point Likert scale. Agents adhering to social distancing norms received a significantly higher favorability rating (mean=3.90) compared to agents aggressively cutting through crowds (mean=2.20) to execute tasks. This difference was statistically significant ($p = 0.012$), highlighting the necessity of incorporating social constraints.

4 CONCLUSION

To address “Access Asymmetry” in Symmetrical Reality, we propose Physical-Social Benchmark, a virtual verification mechanism enforcing rigid physical and social constraints. Empirical results demonstrate that this framework effectively filters unsafe behaviors while catalyzing the emergence of collaboration. Our user study validates the importance of incorporating social constraints for Human-AI coexistence. By ensuring that only agents capable of respecting physical boundaries and social norms traverse the virtual-physical divide, our approach constructs a cornerstone for enduring human-machine trust and symbiosis. Future work targets diverse scenarios for generalizability.

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